

A Project Presentation on

AI SHOP SYNC

Presented By

Priyanshi Tripathi (2202220100137)

Anjali Kushwaha (2202221530004)

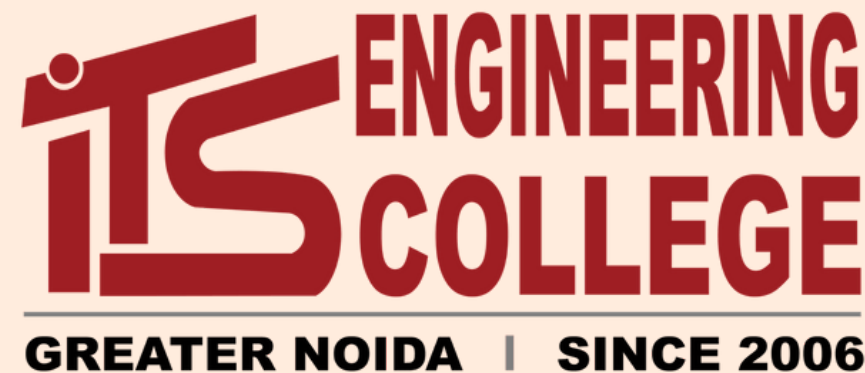
Divya Arya (2202221530019)

Sahil Gupta (2202220100148)

Session 2025–26

Guided By

Ms. Akansha Sharma (Asst. Prof. CSE)



Department of Computer Science & Engineering

I.T.S Engineering College , Greater Noida

Dr. A.P.J. Abdul Kalam Technical University, Lucknow, Uttar Pradesh

ITS ENGINEERING COLLEGE

AI SHOP SYNC



Priyanshi Tripathi (2202220100137)
Anjali Kushwaha (2202221530004)
Divya Arya (2202221530019)
Sahil Gupta (2202220100148)

Guided by :
Ms. Akansha Sharma
Assistant Professor - CSE

Introduction

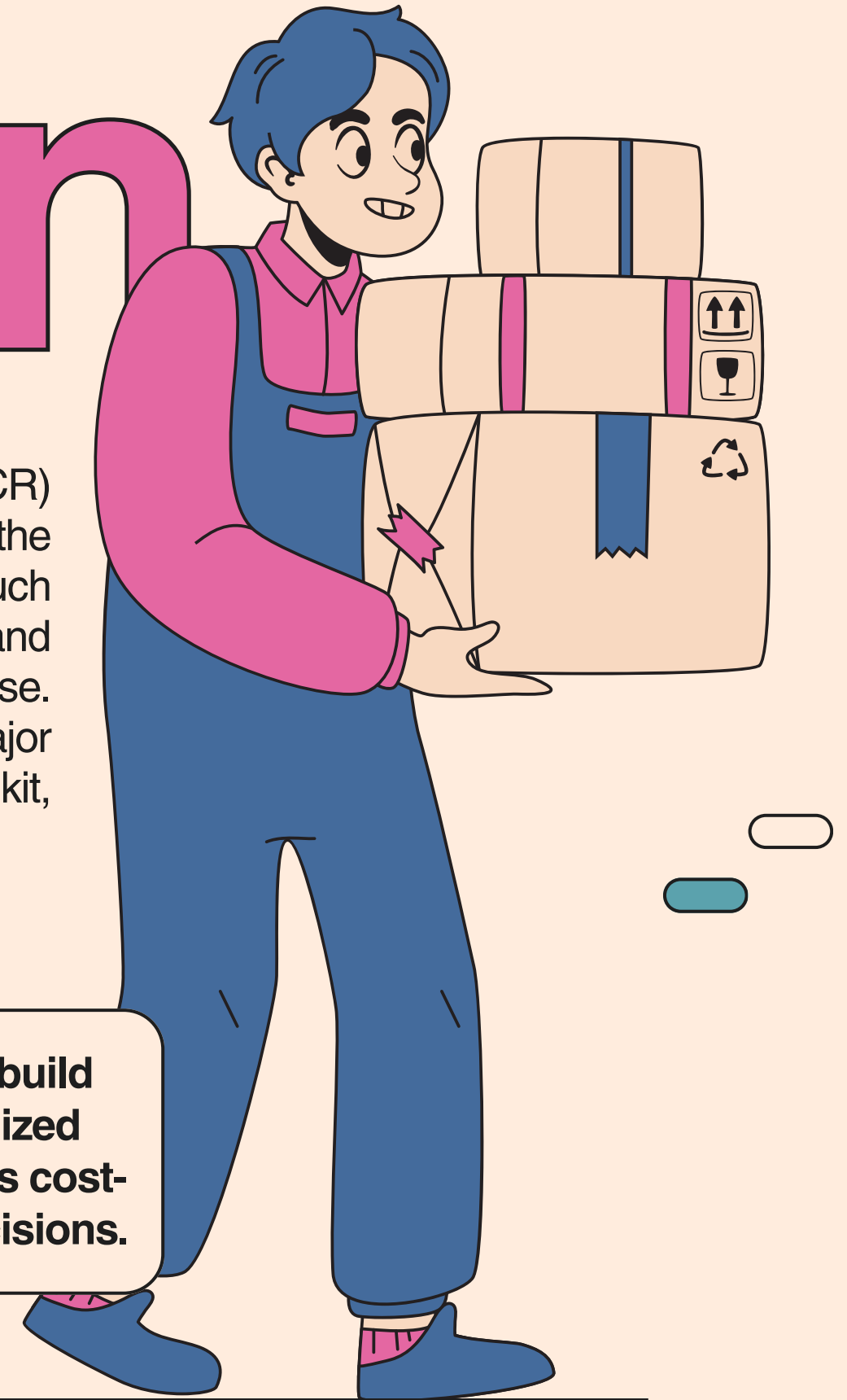
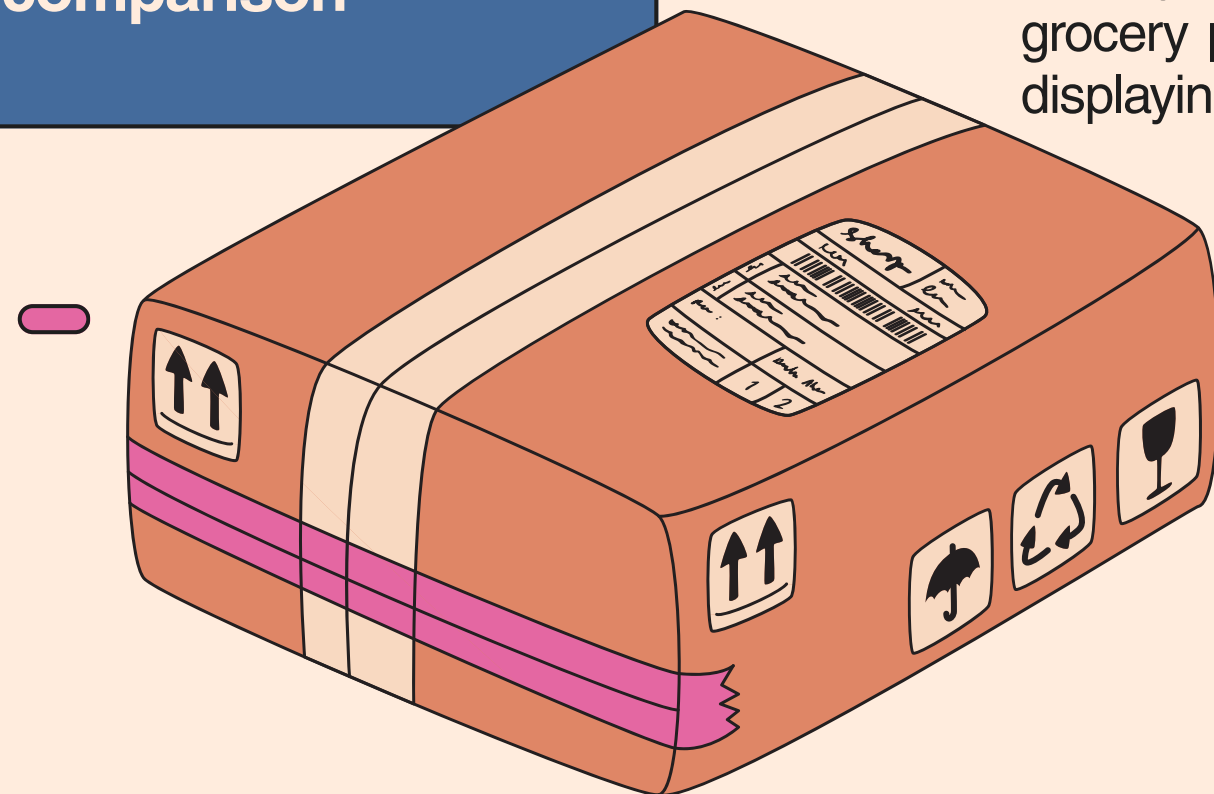
Let's take this forward

Introduction

AI Shop Sync is a mobile-first solution designed for real-time super-market product price comparison

It leverages Optical Character Recognition (OCR) technology to scan product packaging through the user's smartphone camera. Key product details such as name, weight, and MRP are extracted and matched against a standardized product database. The system fetches up-to-date prices from major grocery platforms like Instamart, Zepto, and Blinkit, displaying them directly within the app.

Users can conveniently build and manage a personalized shopping cart that enables cost-optimized purchasing decisions.



Why is it Required?

Problem Statement

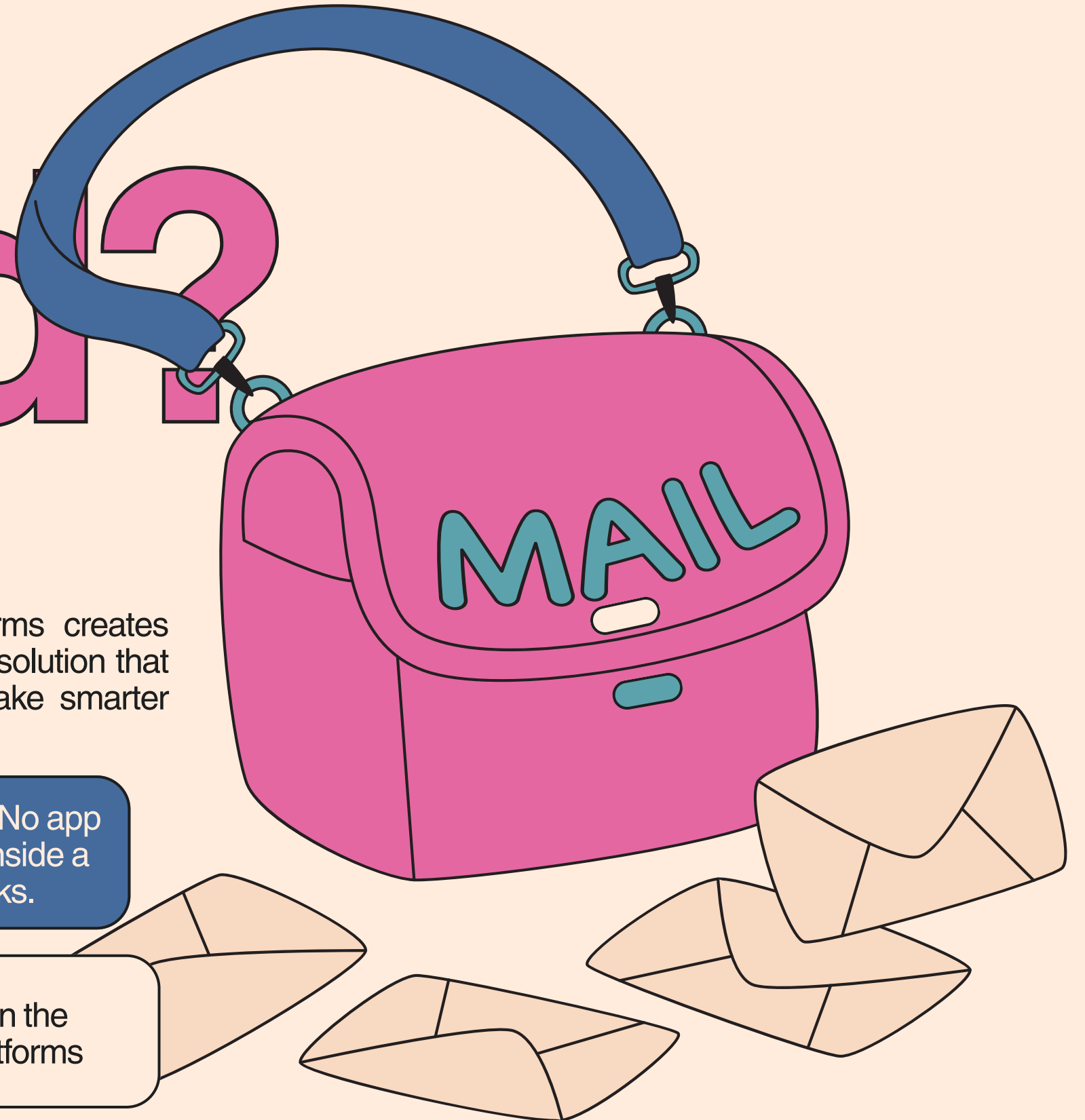
“The increasing variation in product prices across different online grocery platforms creates confusion and higher costs for customers. Our project provides a mobile-based OCR solution that enables real-time product recognition and price comparison, helping customers make smarter purchasing decisions.”

Rising Consumer Costs : Price variations across apps make manual checking difficult.

Lack of Instant Comparison: No app currently allows live scanning inside a supermarket for price checks.

Convenience & Speed : Reduces manual searching and decision-time.

Cart Optimization : Helps users plan the cheapest shopping basket across platforms



Progress in Proposed Work

Activities to be done in order to
complete the project

Progress in Proposed Work

- ☒ Mobile Frontend Development
- ☒ Integration of OCR
- ☒ Backend Development & Integration
- ☐ Web Scraping & Real Time Prices
- ☐ Price Optimization & Cart Handling

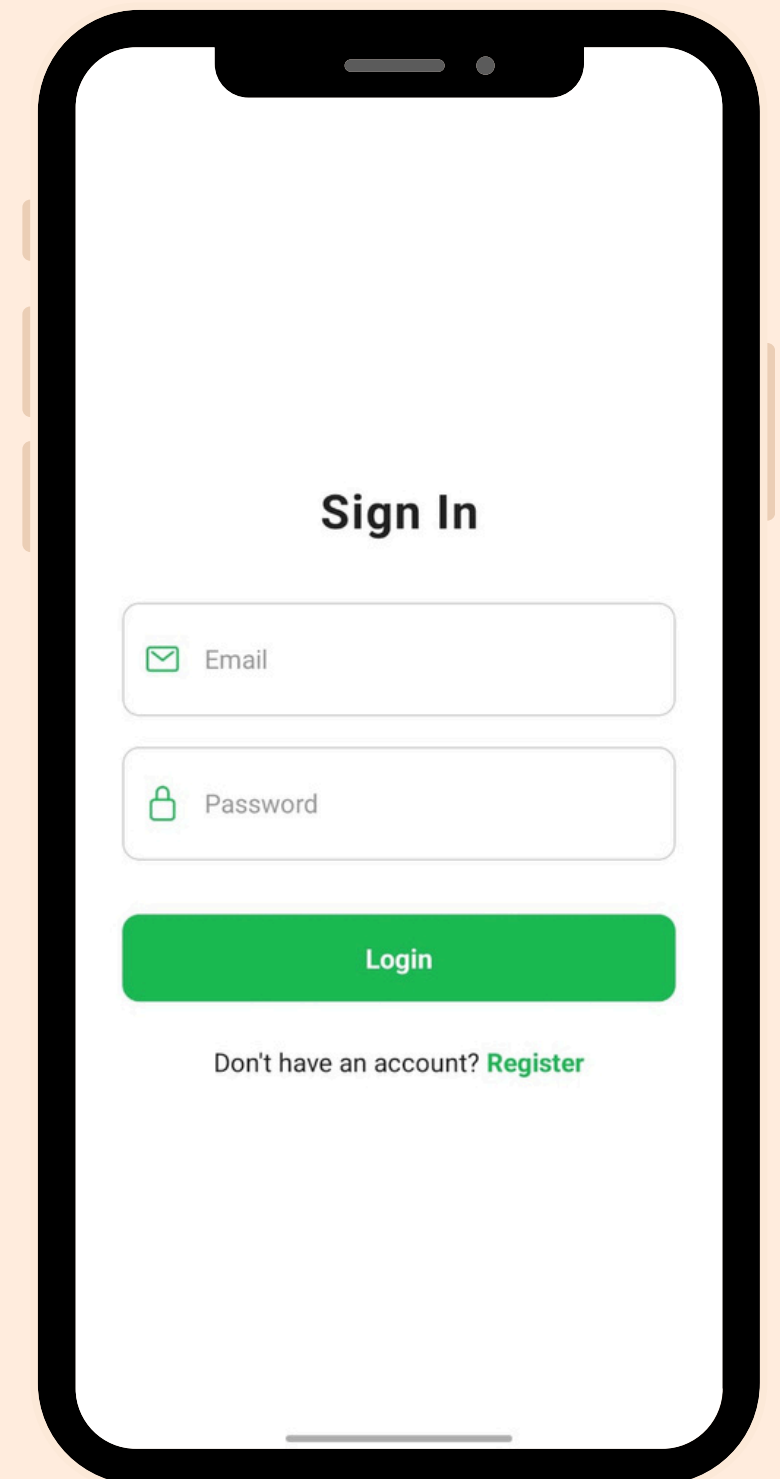


Mobile Front End Development

**Module that covers everything present
on the Mobile Screen**

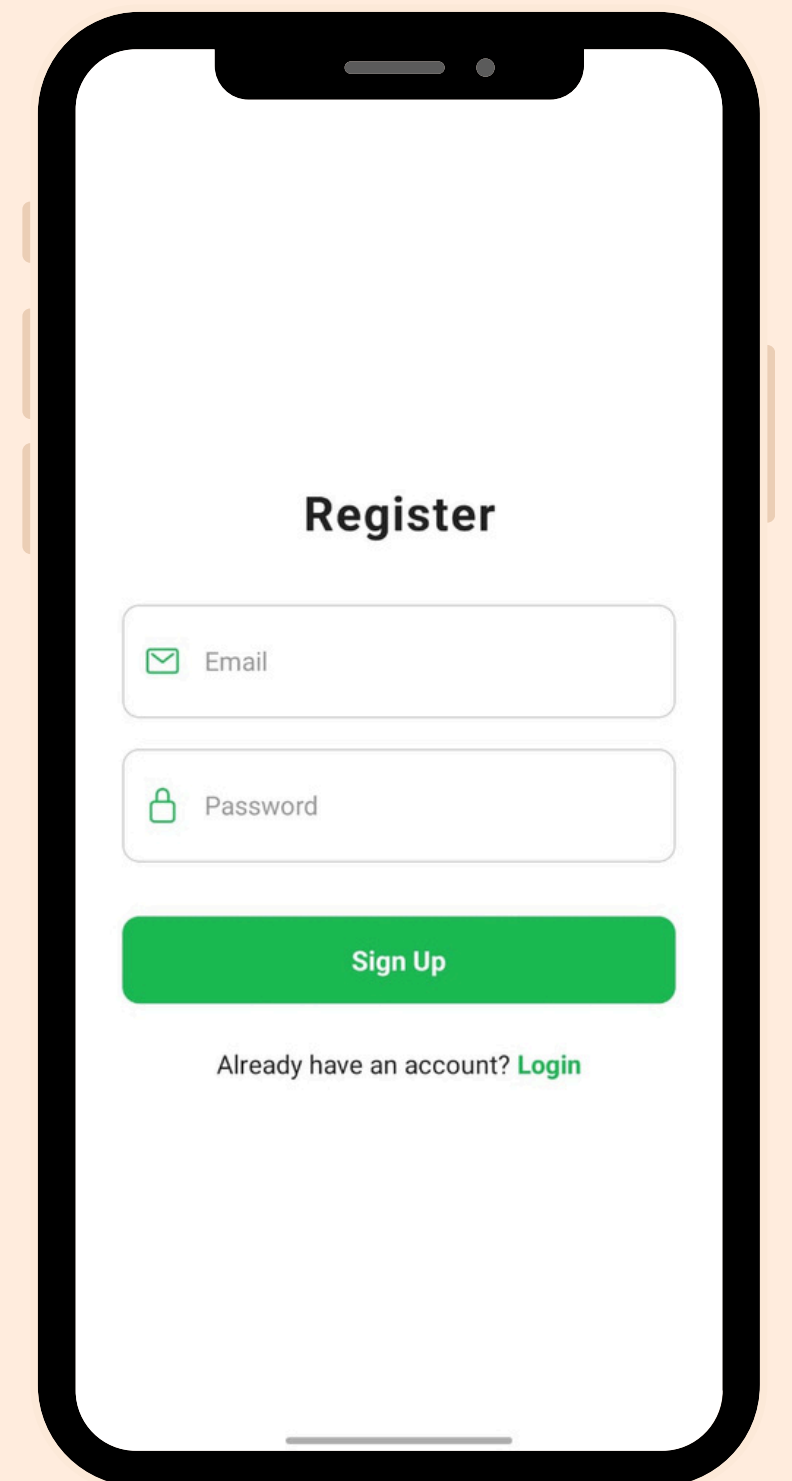
Mobile Front End Development

Sign In Screen - User login interface with email and password fields to authenticate existing users and access their grocery shopping experience.



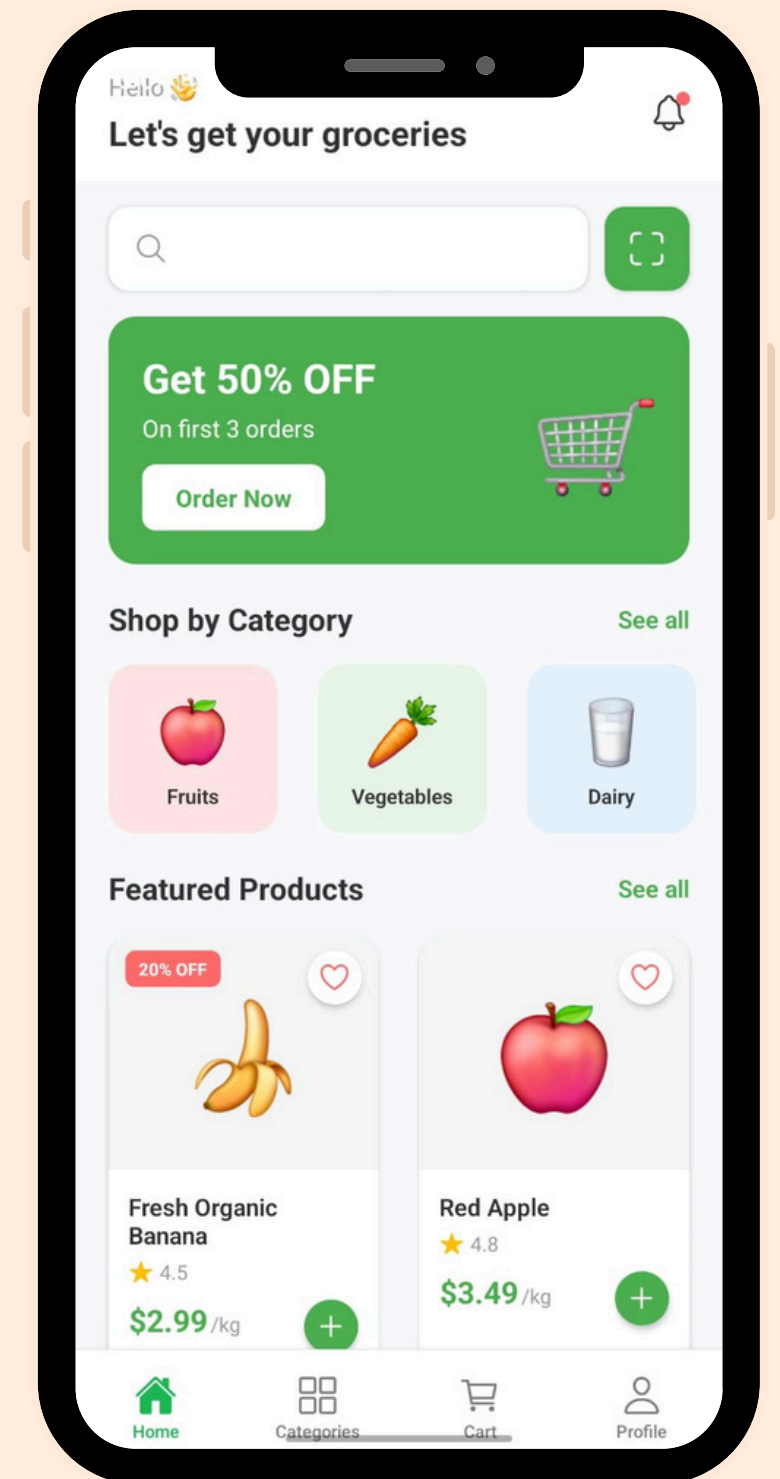
Mobile Front End Development

Register Screen - User sign-up interface with email and password fields to create a new account for accessing the platform.



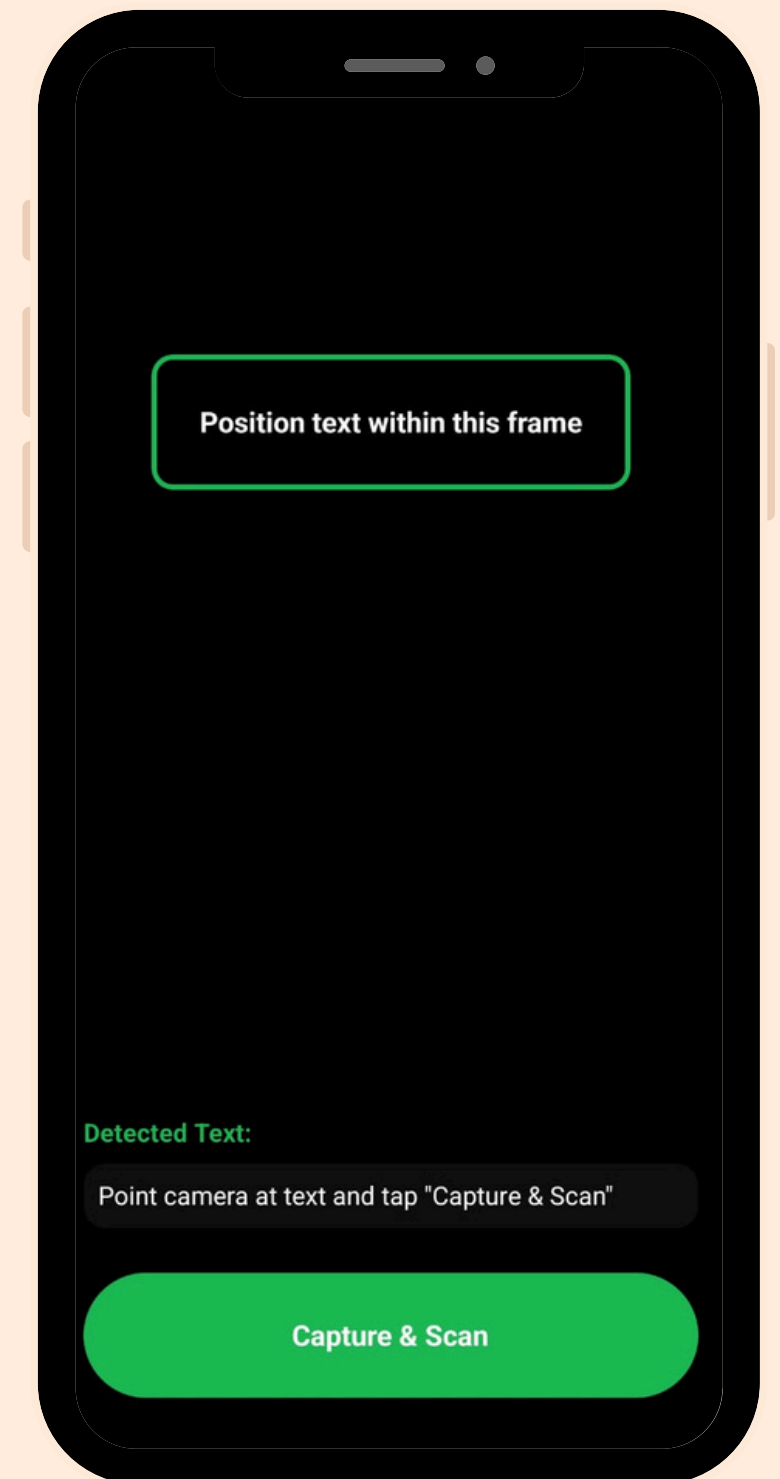
Mobile Front End Development

Home/Landing Screen - Main dashboard showcasing the app's grocery delivery service with promotional banners, category shortcuts, and featured products with discounts.



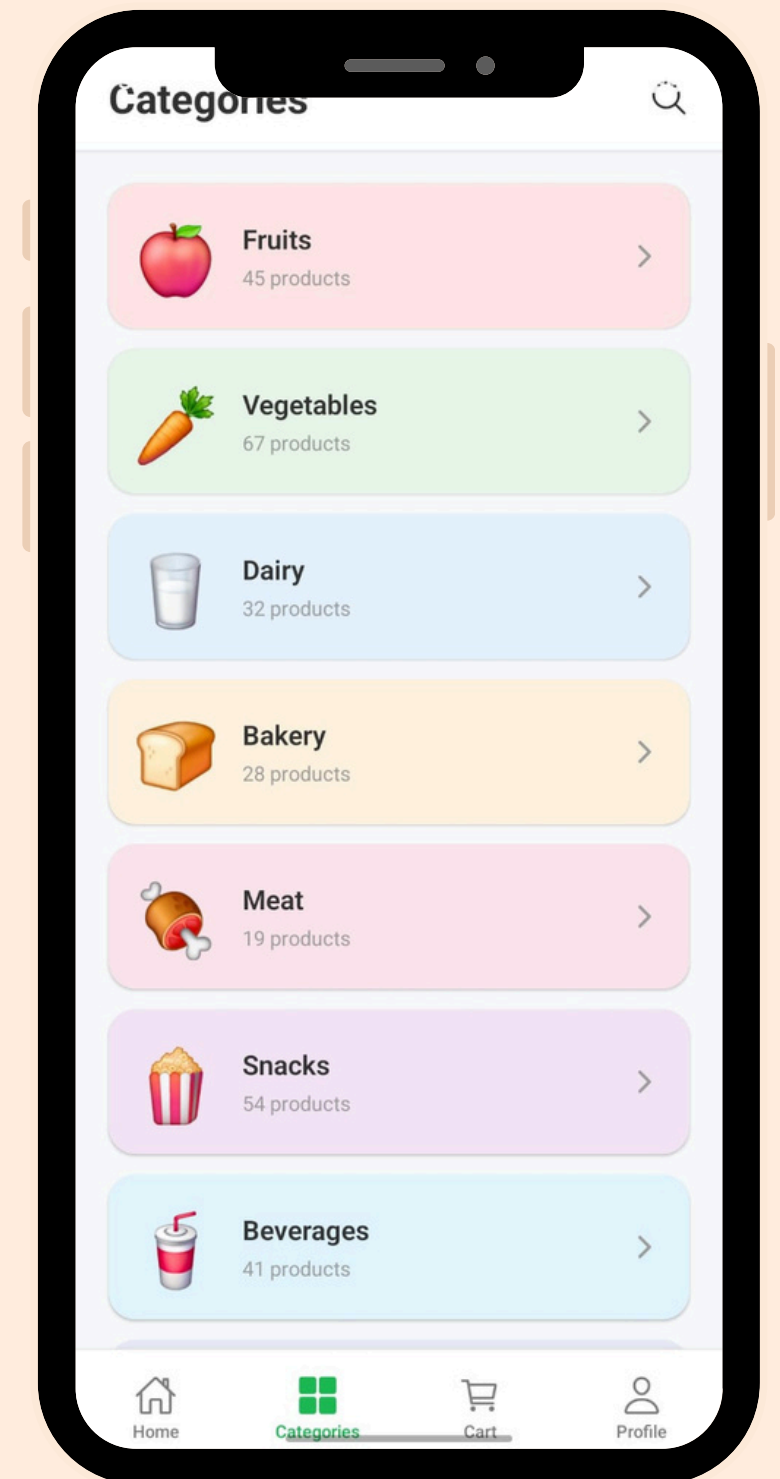
Mobile Front End Development

OCR/Scan Feature Screen - Real-time text recognition functionality using ML Kit that allows users to capture and scan text from physical items directly into the app and then sending that to the server for further elimination of false positives and analysis.



Mobile Front End Development

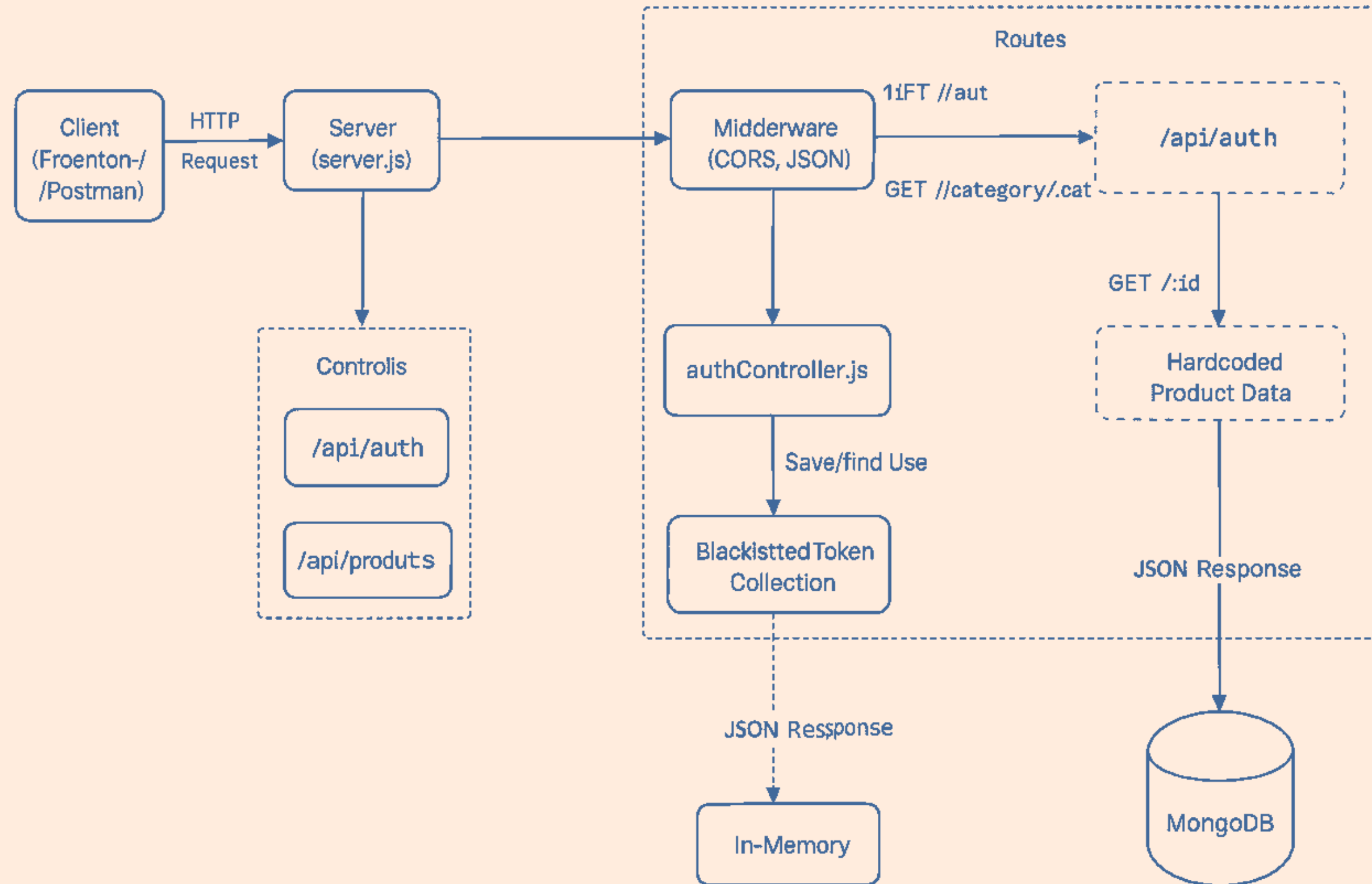
Cart/Checkout Screen - Shopping cart summary displaying added items with quantity controls, pricing breakdown (subtotal, delivery fee, discounts), and a proceed-to-checkout button to complete the purchase.



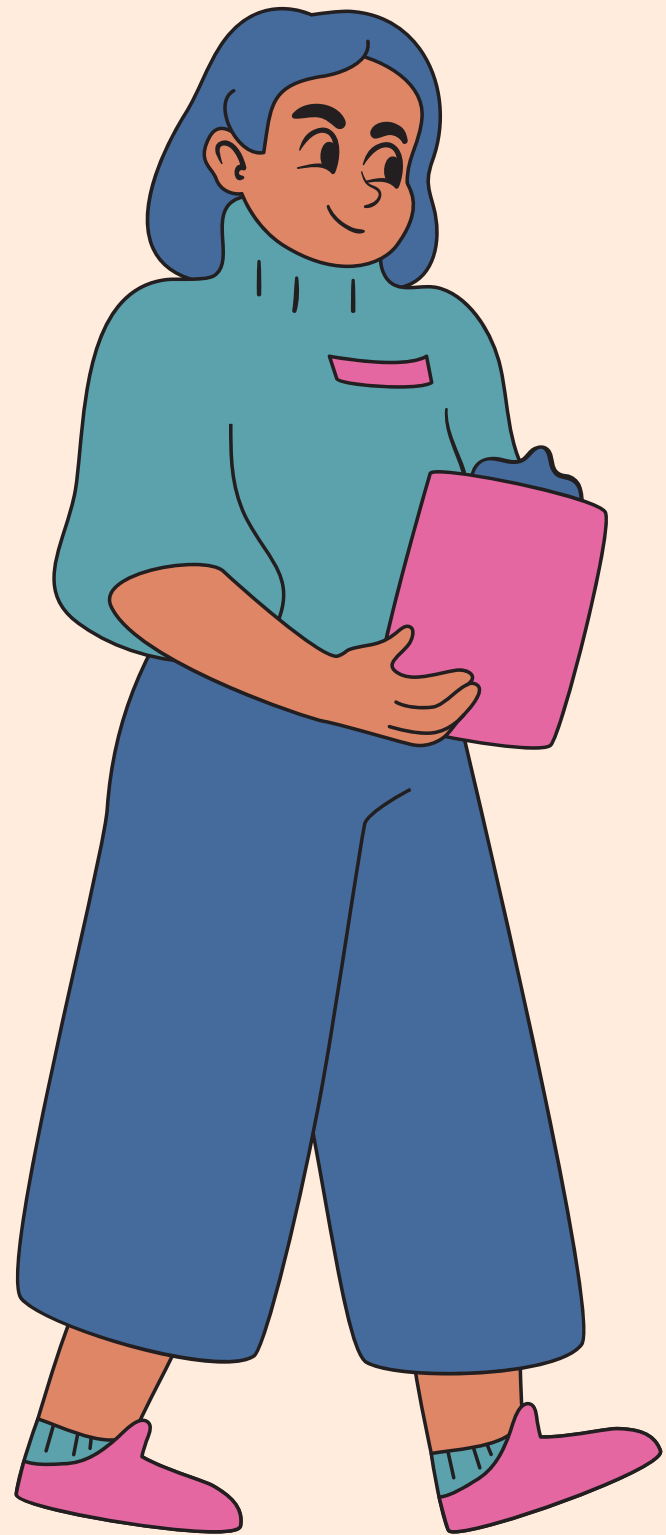
Back End Development

Module that manages everything by
being present in the center.

Back End Development



Back End Development



- **Authentication (JWT & Blacklisting)** - The system uses JWT authentication managed by `authController.js`, which verifies user identity and issues secure access tokens. It enhances security by storing invalidated tokens in a `BlacklistedToken Collection` (synced with In-Memory cache) to prevent reuse after logout.
- **OCR POST Route** - A dedicated POST endpoint captures both the raw image file and the pre-scanned text from the React app, passing them through middleware for validation. This route handles multipart/form-data to securely receive the visual evidence alongside the client-side data payload.
- **Backend Python OCR Processing** - The Node.js server triggers a separate Python script to re-process the uploaded image using advanced OCR libraries (like Tesseract or EasyOCR). This server-side step acts as a verification layer, ensuring the data is accurate by comparing backend results with the mobile app's initial scan before database storage.

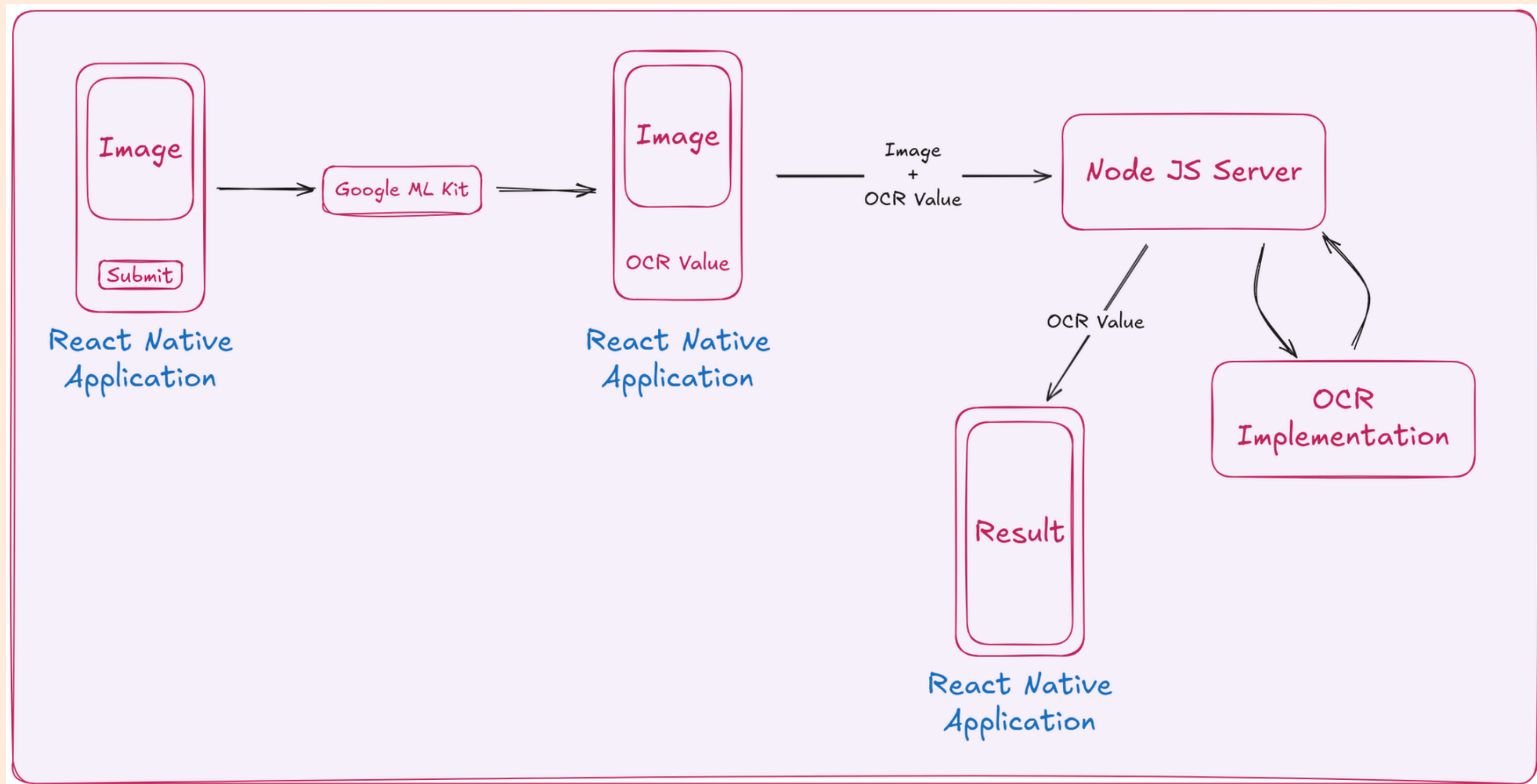
Integration of OCR

**Module that makes the app -- Real World
User Interactive**

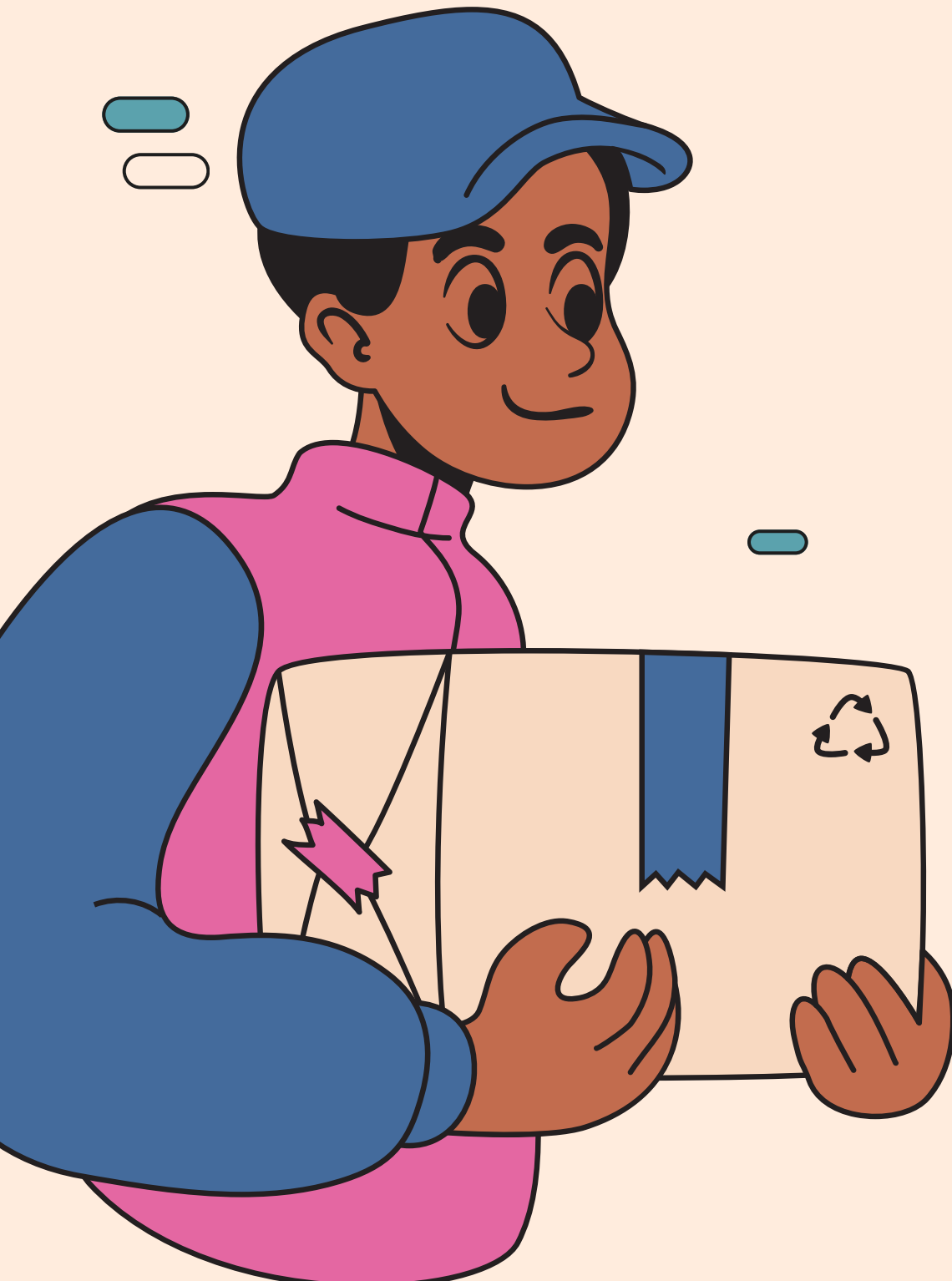
Integration of OCR

OCR Library	Advantages	Drawbacks
PaddleOCR	Fast, lightweight, multilingual support, great for mobile	Less accurate than Tesseract on complex layouts, newer ecosystem
Tesseract OCR	Mature, industry-standard, highly accurate, open-source	Slower processing, high memory footprint, needs preprocessing
EasyOCR	Easy to use, supports 80+ languages, good accuracy, GPU acceleration	Heavier dependencies, slower on CPU, requires download on first use

Integration of OCR



Integration of OCR



Step 1: Client-Side OCR (React Native) - Google ML Kit processes image in real-time on the mobile device, providing instant user feedback and initial text extraction.

Step 2: Image Upload to Server - Original image + initial OCR text is sent to backend via POST `/api/ocr` endpoint for server-side verification.

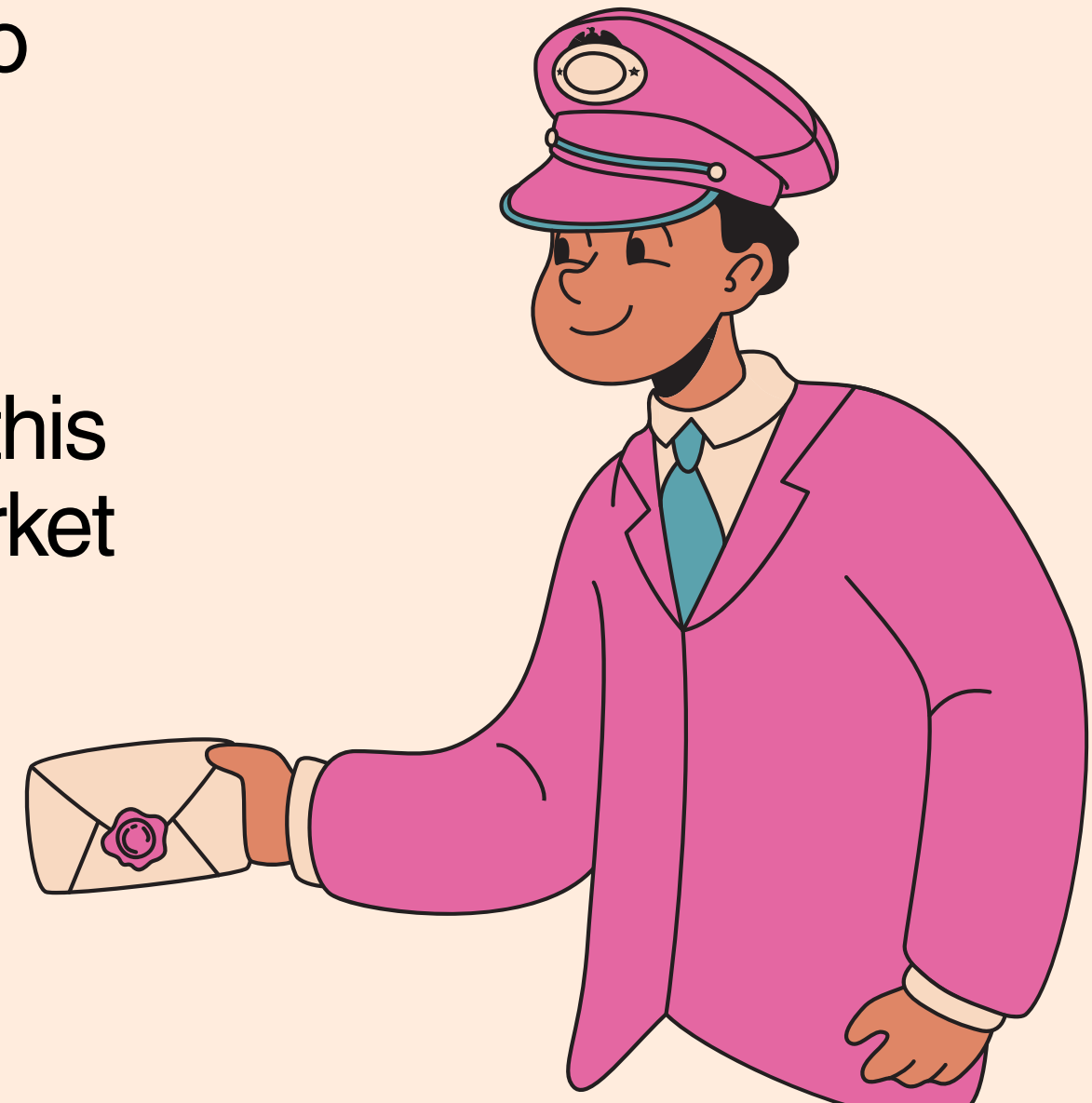
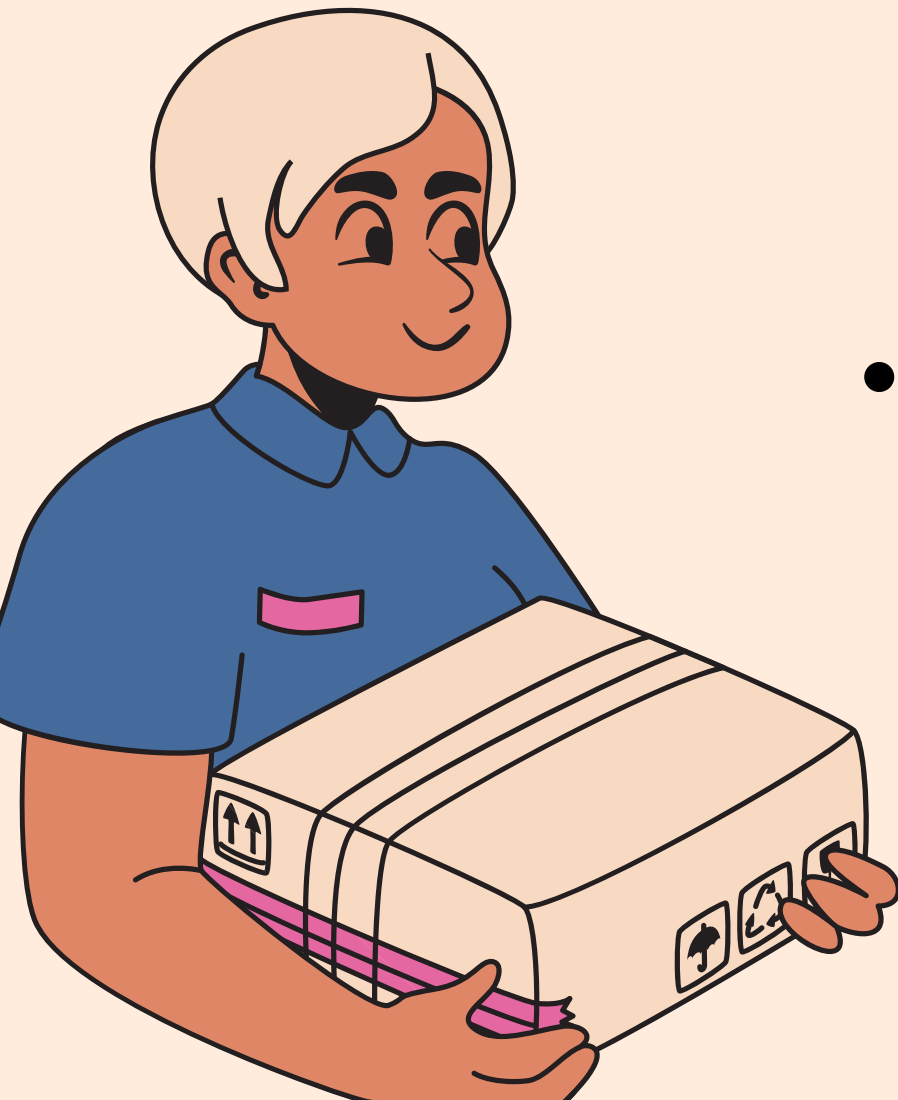
Step 3: Server-Side Re-OCR (Backend) - Tesseract/EasyOCR re-processes the image independently, comparing results with client data to eliminate false positives and ensure accuracy.

Future Progress

**Module that makes the app -- Real World
User Interactive**

Web Scraping & Real-Time Prices

- Develop an automated web scraping tool to monitor competitor pricing and promotions across e-commerce platforms in real-time.
- Create a dynamic pricing model that uses this data to adjust product prices based on market trends, ensuring a competitive edge.



Price Optimization & Cart Handling

- Smart Product Matching (NLP): Uses text similarity algorithms to automatically link identical products across different grocery apps (e.g., matching "Amul Butter 500g" across Blinkit, Zepto, and Instamart).
- Best Basket Recommendation: A simple ranking algorithm that calculates the total cart cost for every app and predicts the "Cheapest Overall Option" for the user's specific list of items.



Thankyou!!!